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Paper 1: AI for NONMEM Coding in Pharmacometrics Research and Education: Shortcut or Pitf

Bibliographic Information

- **Title:** AI for NONMEM Coding in Pharmacometrics Research and Education: Shortcut or Pitfall?
- **Authors:** Zheng W, et al.
- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2025
- **DOI:** 10.1002/psp4.70125

Paper Type: METHODOLOGY

Quick Take

This study benchmarks seven OpenAI LLMs for NONMEM code generation using a standardized rubric and an optimized prompt. o1 and gpt-4.1 achieve perfect scores with the prompt, but complex models still require expert review.

Executive Summary

This study evaluates seven OpenAI LLMs for generating NONMEM code across 13 pharmacometrics tasks, using a novel three-level scoring rubric. Without optimization, performance varied widely; with an optimized prompt, o1 and gpt-4.1 achieved perfect scores on all tasks. The authors highlight that while LLMs can produce structurally correct code for basic models, complex models (e.g., indirect response, TMDD) still require expert oversight. The work provides a benchmark and practical prompt for AI-assisted NONMEM coding.

Scientific Context & Motivation

NONMEM coding is a barrier for many pharmacometricians, especially those without programming experience. Prior studies (e.g., gpt-4.0, Gemini) showed limited accuracy. This work addresses the need for systematic evaluation and prompt engineering to improve LLM performance in pharmacometrics.

✂ Methodological Snapshot

The authors developed a three-level scoring rubric (max 6 points) to evaluate NONMEM code accuracy: Level 1 checks for essential control stream blocks, Level 2 verifies THETA-ETA relationships, Level 3 ensures correct subroutine usage per NONMEM 7.4. They tested seven OpenAI LLMs on 13 tasks (basic to complex PK/PD models) with and without an optimized prompt that embeds rubric criteria. Scoring was done by three annotators with pharmacometrics expertise.

⊠ Statistical Framework

Not applicable — the study evaluates code accuracy, not statistical inference.

Estimator Behavior

Not applicable — the study evaluates code accuracy, not estimator properties.

Validation Design

13 NONMEM coding tasks (1- to 3-compartment PK, TMDD, indirect response, etc.) were evaluated by three annotators using a 3-level rubric. Performance was compared across seven LLMs with and without an optimized prompt. Wilcoxon matched-pairs signed rank test was used for statistical comparison. No model execution was performed; only code structure was assessed.

Comparison to Alternatives

The study compares seven OpenAI LLMs (gpt-4.1-mini, gpt-4.1-nano, gpt-4.1, gpt-4o-mini, gpt-4o, o1, o3-mini) against each other using a standardized rubric. o1 and gpt-4.1 outperform others, especially with the optimized prompt. No comparison to non-OpenAI models (e.g., Gemini, Claude) is made, limiting generalizability.

Implementation Guidance

The optimized prompt (provided in Data S1) should be used as a template for NONMEM code generation. Users must review and adapt code for dataset alignment and advanced features. The scoring rubric can serve as a checklist for code quality. Computational cost is minimal (API calls).

⊠ Key Findings

OpenAI o1 and gpt-4.1 achieved perfect scores on all 13 tasks with the optimized prompt. Without optimization, median scores ranged from 2 to 5. The optimized prompt improved performance across all models, with greatest gains in complex tasks (indirect response, absorption lag time). The scoring rubric (3 levels, max 6 points) provides a standardized evaluation framework.

Strengths & Limitations

Strengths

- Standardized, reproducible scoring rubric.
- Comprehensive task set covering basic to complex PK/PD models.
- Practical optimized prompt provided for immediate use.
- Clear discussion of educational implications and risks.

Limitations (Acknowledged by Authors)

- Models were not executed; only code structure was evaluated.

- No sample datasets were generated for testing.
- Scoring rubric may not capture all nuances of NONMEM code correctness.

Limitations (Expert Review)

- Single annotator consensus without formal inter-rater reliability metrics.
- No comparison to non-OpenAI models.
- Optimized prompt may overfit to the specific rubric and tasks.

Generalizability

Limited to OpenAI models; results may not extend to other LLMs (e.g., Gemini, Claude). Tasks are representative but not exhaustive.

Figures & Tables

- **Figure 1:** Boxplots of LLM performance scores (0–6) for 13 tasks, with and without optimized prompt.
 - *Significance:* Shows o1 and gpt-4.1 achieve highest median scores; optimized prompt dramatically improves all models.
 - **Figure 2:** Task-specific score improvements with optimized prompt across all LLMs.
 - *Significance:* Demonstrates that complex models (indirect response, lag time) benefit most from prompt optimization.
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Code & Reproducibility Assessment

The authors provide all generated NONMEM codes and prompts in Data S1, but do not include executable code or datasets for running the models.

Supplementary Materials

Data S1 contains all generated NONMEM codes and the optimized prompt.

Future Directions

Future work should include running the generated models with simulated datasets to assess execution accuracy, extending evaluation to non-OpenAI LLMs, and developing domain-specific fine-tuned models for pharmacometrics.

Expert Commentary

The study is a timely and practical benchmark for AI-assisted NONMEM coding. The optimized prompt is a valuable contribution, but the lack of model execution and reliance on a single rubric limit the assessment of real-world usability. Future work should include runtime validation and broader model comparisons.

Bottom Line

LLMs, particularly OpenAI o1 and gpt-4.1, can generate accurate NONMEM code for basic PK/PD models when guided by an optimized prompt, but expert review remains essential for complex models and dataset-specific adjustments.

☒ Figures

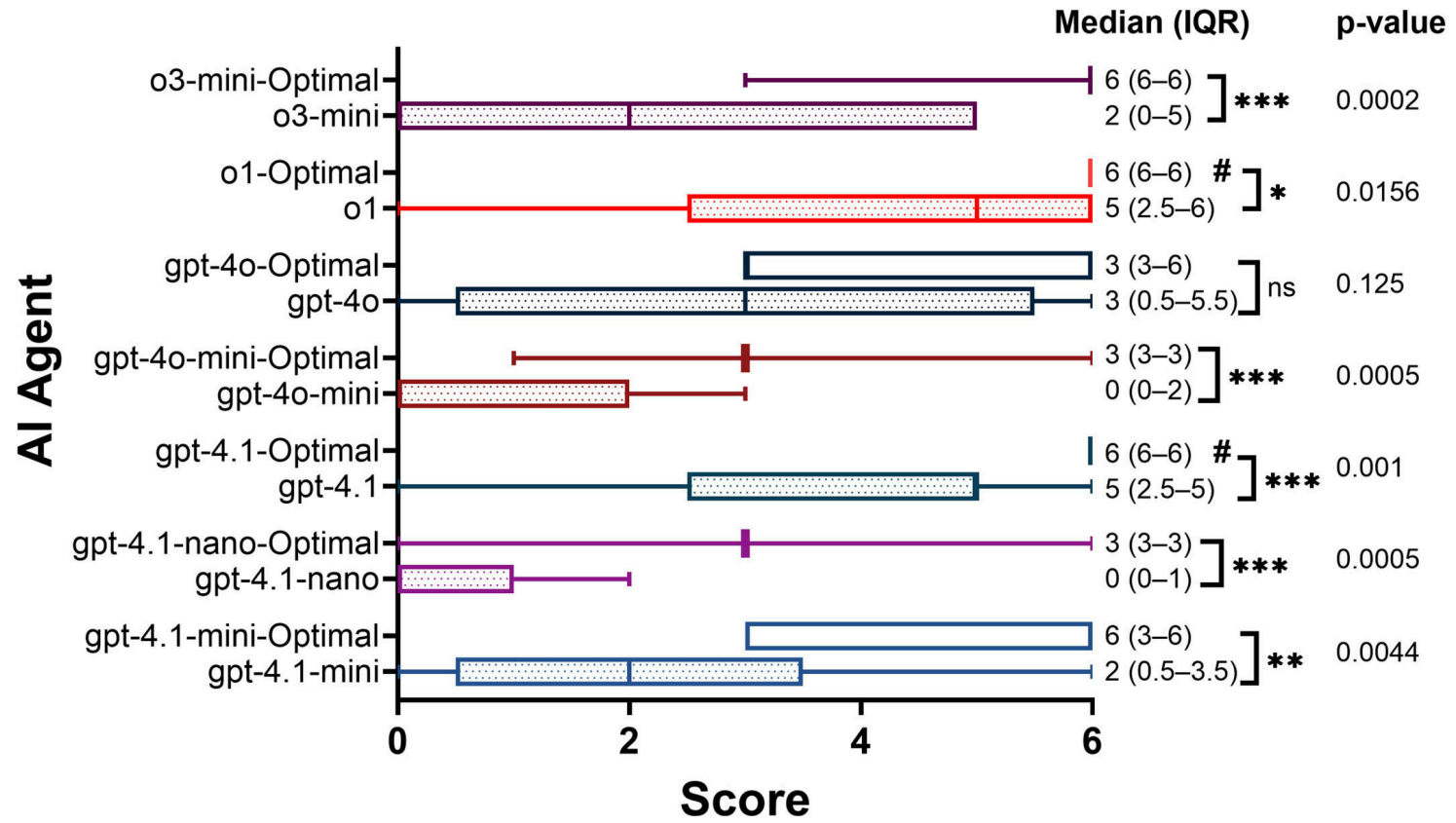


Figure 1: Summary of LLM performance. NONMEM codes generated by different LLMs for 13 tasks were evaluated using a predefined scoring rubric. Each boxplot displays the med

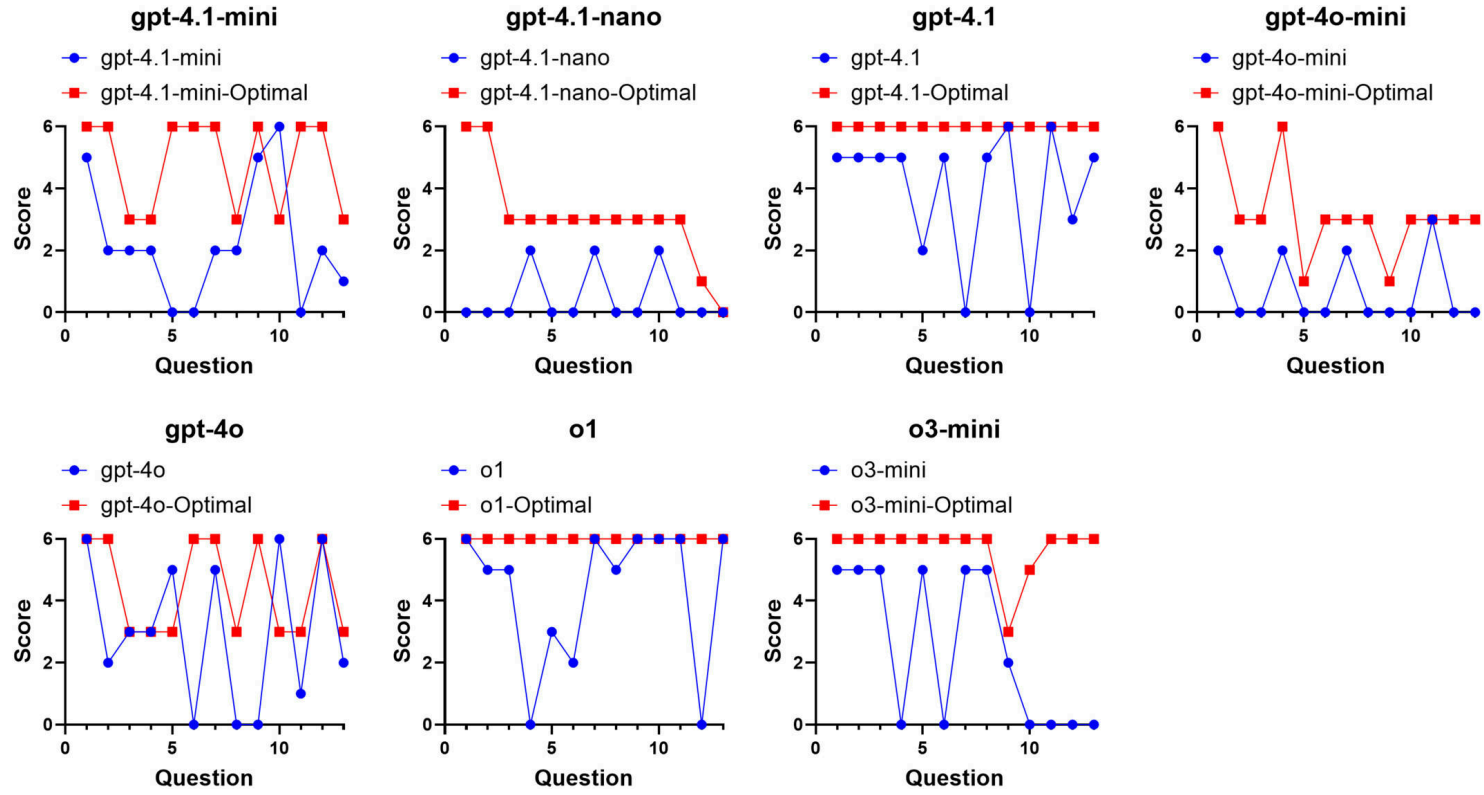


Figure 2: LLM performance improved with optimal algorithm in writing NONMEM codes. The score of each question was compared between original LLM and LLM-optimized using rub